Md Ahsanul Haque, Ph.D.
MAT-120

Final Exam

Date: May 11, 2026

Name

DISTRIBUTION OF MARKS		
Question No.	**Full Marks**	**Obtained Marks**
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
Total	50	

Instructions:

- Write your full name (first **and** last name) **inside** the provided box.
- This assignment contains 6 problems. It is **YOUR responsibility** to check that your copy has all 6 problems.
- Answer **5** questions. If you answer 6, the first 5 will be graded.
- Use correct mathematical notation and show all logical steps.

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1. (10 points) Solve the following problems
   (a) A spherical snowball is melting so that its radius is decreasing at a constant rate of 1.5 cm/min.
      i. Determine the rate at which the volume of the snowball is decreasing when the radius is 4 cm.
      ii. Determine the rate at which the surface area is decreasing at that same moment.
   (b) A street light is mounted at the top of a 6 m tall pole. A person 1.5 m tall walks away from the pole with a speed of 2 m/s along a straight path.
      i. How fast is the length of the person's shadow increasing when the person is 8 m from the pole?
      ii. How fast is the tip of the shadow moving away from the pole at that moment?

Extra Space

2. (10 points) Evaluate the following limits

   (a) $\lim_{x\to 0} \dfrac{e^{3x} - 1 - 3x}{x^2}$.

   (b) $\lim_{x\to \infty} \dfrac{\ln(x^5)}{x}$.

   (c) $\lim_{x\to 0^+} x \ln x$.

   (d) $\lim_{x\to 0} \dfrac{x - \sin x}{x^3}$.

Extra Space

3. (10 points) Consider the polynomial function $f(x) = 2x^3 - 9x^2 + 12x + 1$.
   (a) Find the intervals where $f(x)$ is strictly increasing and strictly decreasing.
   (b) State the intervals of concavity.
   (c) Find the absolute maximum and absolute minimum values of $f(x)$ on the interval $[0, 3]$.
   (d) What's your observation on the absolute extrema if we consider the domain of f?
   (*Hint: You don't need any calculation here. Just look at the function!*)

Extra Space

4. (10 points) Solve the following problems
   (a) Find the following derivatives:
      i. $g(x) = \cos^{-1}(2x - 1)$.
      ii. $h(x) = \dfrac{x^x \sqrt[3]{x+1}}{e^{2x}}$.
   (b) Approximate the area under the curve $f(x) = x^3 + 2$ on the interval $[0, 2]$ using a right-endpoint Riemann sum with $n = 4$.
   (c) Use sigma notation to find the following sum (Show your work!): $1^2 + 2^2 + 3^2 + \cdots + 20^2$

Extra Space

5. (10 points) Evaluate the indefinite integrals

   (a) $\int x(x^2+1)^3\,dx.$

   (b) $\int \dfrac{e^x}{1+e^x}\,dx.$

   (c) $\int \dfrac{\cos(\ln x)}{x}\,dx.$

   (d) $\int \dfrac{x^3}{1+x^8}\,dx.$

Extra Space

6. (10 points)
   (a) Use the Fundamental Theorem of Calculus to solve the following:
      i. Find $F'(x)$ when $F(x) = \int_0^x \sqrt{4+t^5}\, dt$.
      ii. Evaluate exactly: $\int_0^{\pi/6} \sec^2(x)\, dx$.
   (b) Evaluate the following definite integrals
      i. $\int_0^{\pi/4} \sec x \tan x\, dx$.
      ii. $\int_{\pi/6}^{\pi/4} \csc^2 x\, dx$.

Extra Space